BES Report Draft 2

Over recent years, the scientific community has been facing a replication crisis, with growing numbers of studies failing to reproduce previously published findings @klugkist_volker. In this light, research synthesis methods that aggregate evidence from multiple replication studies, have gained more popularity @vanlissa. One of the most well known research synthesis methods is meta-analysis, in which results from multiple studies are pooled on the level of effect sizes, increasing the power of the individual studies @klugkist, @vanlissa. Especially when using random-effects meta-analysis (RMA), which accounts for random between-study variation, studies do not need to be exact replications, making it a good way to deal with the replicaiton crisis.

However, because results are pooled at the effect size level, a drawback of meta-analysis is that the effect sizes and methodologies of the synthesized studies need to be similar/comparable. So even when studies are theoretically similar and focus on the same underlying hypothesis, meta-analysis may not always be able to synthesise their results if their effect sizes differ @volker_klugkist. An alternative synthesis method that does allow synthesis on the hypothesis level, rather than the effect size level, is Bayesian Evidence Synthesis (BES) @volker, @vanlissa. Because the method does not pool the data, one disadvantage is that BES does not increase the power, but rather shows whether all studies align in the presence and direction of any effect.

In a simulation study by @vanlissa, BES was shown to be a promising synthesis method. This internship was aimed to replicate this simulation study, and expand it to new scenarios to test whether the very positive image they pained of BES would remain the same. *[mention here that they used an arbitrary threshold or do that later??]*

Bayesian Evidence Synthesis

To understand BES, it is important to first understand how Bayes Factors (BFs) are defined. BFs measure the relative probability of the data under two different hypotheses, H_1 and H_2 @volkerklugkist, @hoijtink2019. This means that an informative hypothesis H_i is always tested against an alternative hypothesis H_a. Three hypotheses that are often used as alternatives are (1) the unconstrained hypothesis, H_u, that includes all possible values within the parameter space, (2) the complement of the informative hypothesis, H_c, that includes the set of all values that are not in H_i, and (3) the null hypothesis, H_0.

The BF is calculated by taking the ratio of the marginal likelihoods of the two hypotheses $P(E|H_1)$ and $P(E|H_2)$:

$$BF_{ia} = \frac{P(D|H_i)}{P(D|H_a)}. \tag{1}$$

When the BF_{ia} is larger than 1, this means there is more support for H_i, while a BF smaller than 1 implies evidence in favour of H_a. A BF of 3 indicates that there is three times more evidence in favour of H_i than for H_a.

Another way of conceptualising the BF, is in terms of the relative fit and relative complexity of H_i versus H_u:

$$BF_{iu} = \frac{f_i}{c_i}, \tag{2}$$

where the fit denotes the area under the unconstrained posterior curve that agrees with H_i, and the complexity is the area under the unconstrained prior curve that agrees with H_i @hoijtink2019. By definition, Bayes Factors reward hypotheses that fit the data well without being unnecessarily complex, and penalise hypotheses that are either too restrictive (poor fit), or insufficiently restrictive by allowing many values that are not in line with the data (too complex).

In Bayesian Evidence Synthesis, study results are aggregated on the level of Bayes Factors. First, individual BFs are calculated per study. Assuming that all studies provide independent evidence for the hypothesis under consideration, the results can be synthesised by simply taking the product of all individual BFs (hereafter referred to as pBF):

$$pBF_{iu} = \prod_{i=1}^{K} (BF_{iu})^k, \text{ where k = 1, ..., K is the number of studies} \tag{3}$$

This way, BES quantifies the amount of evidence for an overarching hypothesis in each of the studies.

Performance of Bayesian Evidence Synthesis Versus Other Synthesis Methods

Although one of the major advantages of BES is that it is a flexible method that allows synthesis of studies on a conceptual level, it is important to highlight that BES does not pool the results and thus does not increase power to find support for a hypothesized effect, contrasting with other synthesis methods such as Bayesian updating or meta-analysis. @vanlissa performed a simulation study to evaluate the pBF algorithm and compare its performance with other evidence synthesis methods. They contrasted the results of BES, where BFs were calculated using `bain`, to those of RMA, individual participant data (IPD), and vote counting. Specifically, they focused on the sensitivity (detecting an effect when it is truly present in the population, true positives (TP)) and specificity (ability to reject H_i when it is not supported in the population, true negatives (TN)). They used $H_i : \rho > 0.1$, and tested this under different conditions, shown in Table 1.

Condition	Value
Number of Studies	2, 3, 5
Within Study Sample Size	20, 80, 200, 500
True Effect Size	0.1, 0.2
Reliability	0.6, 0.8, 1
Error Corresponding to Reliability	0.81, 0.5, 0

For RMA and IPD, H_0 was rejected if the 90% confidence interval excluded the hypothesised value under H_i. For vote counting, they simply counted the number of significant results, and if H_0 was rejected in the majority of all studies, it was classified as evidence for H_i. For BES, a decision criterion of a pBF larger than 3 was used to classify support for H_i. Using the true population effect sizes corresponding to H_i or H_0, they found that the pBF had relatively high sensitivity and low specificity, compared to the other methods.

Present Study

One drawback of the simulation by @vanlissa is that they used an arbitrary decision criterion to evaluate the pBF. Given that BES is a relatively new synthesis method, it is highly relevant to get a complete picture of its performance under different conditions. Moreover, as @hoijtink argues, unlike p-values, the BF can and should be interpreted on a continuous scale, which will likely give a better overview of the general performance of BES.

This internship was focused on evaluating the performance of BES based on the full distribution of pBF values, to better understand its behaviour across conditions, aiming to identify the parameters that seem to influence the results the most. To better visualise the pBF distributions, all pBF values were log_{10} transformed. In addition, after exploring the pBF distributions, this internship extended the original simulation by also including other effect size types, namely covariances and regression coefficients (β). Additionally, instead of testing only $H_i : \rho > 0.1$, this internship also extended the simulation by including an ordered hypothesis: $H_i : \rho_1 < \rho_2 < \rho_3$ to evaluate the performance of the pBF in this scenario.

`Because the original simulation study focused on TP and TN, this internship mainly explored $pBF_{ic}$ distributions, as in this case either $H_i$ or $H_c$ would be true.` As it is already known that especially in boundary cases, where neither H_i or H_c is true, the performance of BES is not always optimal @elise, this scenario was highlighted in this internship. In this simulation, the boundary case was when $H_i : \rho > 0.1$ was tested against $H_c : \rho \leq 0.1$ and the true effect was $\rho = 0.1$. Given that an estimate can never exactly equal any specific value, in this case, by definition both H_i and H_c are not true. Therefore, in this boundary case, we would expect all pBF values to be centered around 1. This would indicate neutral evidence for either hypothesis.

Simulation 1: Replication / Correlation

`Except for the results of the pBF, the confusion matrix did not completely replicate, which was likely due to slight numerical sensitivity of the evidence methods (see <github.com/lodewijn> for the replication scripts).`

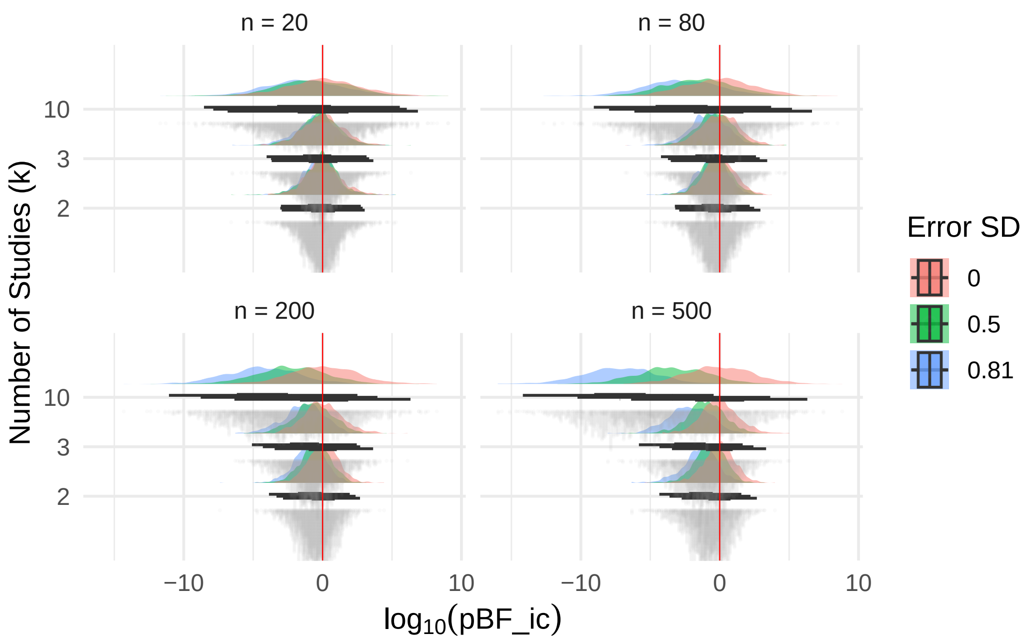


Figure 1: Boundary case, true correlation = 0.1, Hi and Hc are both false. Red line: log(1) = 0, 1000 iterations, Number of studies (k), within study sample size (n) and reliability (errorsd)

In Figure 1, the true effect size (`es`) equals 0.1, which means that the true parameter value is on the boundary of the hypotheses H_i and H_c. Therefore it is expected that the $log_{10}pBF_ic$ would be distributed symmetrically around zero, being unable to provide convincing evidence for either hypothesis.

Especially with increasing `errorsd` values, an interesting pattern emerged. Whereas the $log_{10}pBF_ic$ distributions were symmetrical around zero when the error was 0, at higher error values the $log_{10}pBF_ic$ shifted towards evidence in favour of H_c. If the within-study sample sizes (`n`) are small (e.g., 20 or 50), the shift was less prominent. However, when the within-study sample sizes (`n`) increase (e.g., 200, 500), especially in combination with higher numbers of studies (`k`), H_c was consistently favoured over H_i. This suggests that higher error, especially in combination with many and large studies, will result in a stronger shift towards evidence for H_c.

The parameter of interest in this simulation study is a correlation, implying that the full parameter space is constrained: $\rho \in [-1, 1]$. This means that H_c contains a larger proportion of the full parameter space ($\rho \in [-1, 0.1]$) than H_i ($\rho \in [0.1 - 1]$), raising the question if the distributional shift we observe when es $= 0.1$ can be explained by the fact that the two hypotheses are not equally large in terms of the parameter space. When there's error, will the evidence distribution always shift in favour of the hypothesis that contains the largest parameter space? This was be tested by running the simulation again, while testing the following hypotheses:

- (1b) $H_i : \rho > 0$ versus $H_c : \rho \leq 0$, when es $= 0$, in this case it is expected that at all error values, the distributions will be symmetrical around $log_{10}\ pBF_{ic}$ values of zero, as both hypotheses entail an equally large parameter space.
- (1c) $H_i : \rho > 0.8$ versus $H_c : \rho \leq 0.8$, when es $= 0.8$, in this case it is expected that with higher error values, there will be an even more dramatic shift towards negative $log_{10}\ pBF_{ic}$ values, indicating evidence for H_c.
- (1d) $H_i : \rho > -0.8$ versus $H_c : \rho \leq -0.8$, when es $= -0.8$, in this case it is expected that with higher error values, there will be shift towards positive $log_{10}\ pBF_{ic}$ values, indicating evidence for H_i, as now this hypothesis will entail the largest proportion of the parameter space.

Constrained Parameter Space and Complexity

The simulated results showed exactly what we expected: when the reliability was low (high error), the $log_{10}\ pBF_{ic}$ always tended towards evidence for the hypothesis with the largest parameter space. When the boundary value was more extreme (e.g., 0.8 or -0.8) the shift in evidence was also more extreme (see Figure 2 for an example). Moreover, in simulation 1b where both hypotheses had equal parameter spaces ($H_i : \rho \in [0, 1]$ and $H_c : \rho \in [-1, 0]$) it stood out that all pBF distributions were centered nicely around a $log_{10}\ pBF_{ic}$ of zero.

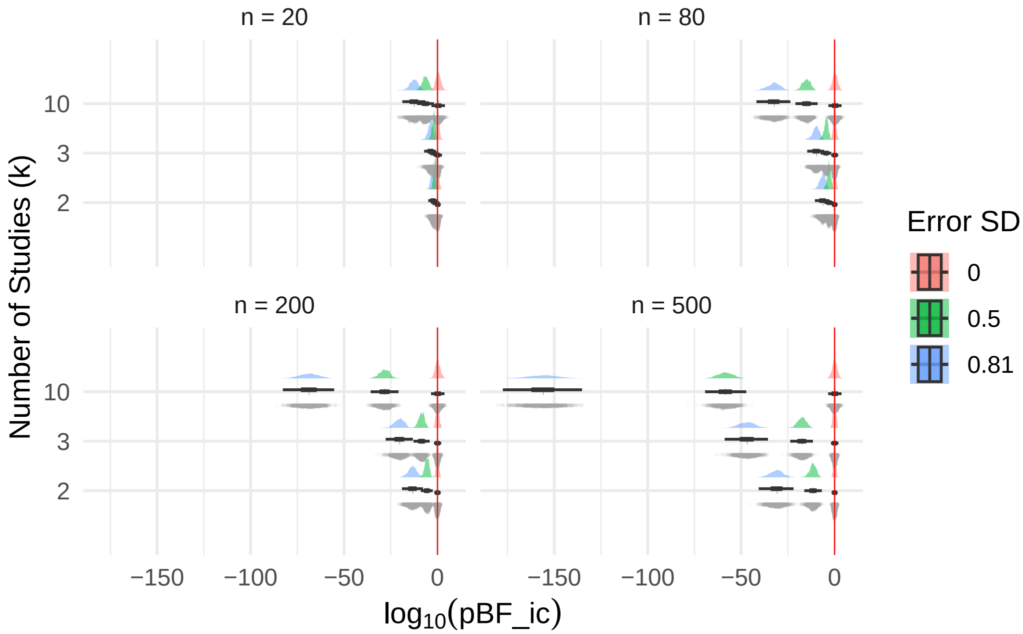


Figure 2: Boundary case, true correlation = 0.8, Hi and Hc are both false. Red line: log(1) = 0, 1000 iterations, Number of studies (k), within study sample size (n) and reliability (errorsd)

This suggests that the constrained parameter space of correlations may have played a role. To explain why error seems to affect this shift in evidence, we looked at the way `bain` defines its parameter prior. According to @hoijtink2019, the prior distribution for the population correlation is chosen in such a way that it accurately reflects the complexity of each hypothesis, and is specified as:

$$\rho_g \sim N(\rho_B, \frac{1}{bg} \cdot \frac{\hat{\sigma}^2}{Ng}), \tag{4}$$

where:

- ρ_B is the boundary value of the two hypotheses (e.g., 0.1)
- $\frac{1}{bg} = \frac{1}{Ng}$ is the fraction of information used from the data (default = 1)
- $\hat{\sigma}^2 = SE_r^2$ is the estimated sampling variance of the correlation
- Ng is the group sample size

This means that the prior is defined as:

$$\rho_g \sim N(\rho_B, \frac{Ng}{1} \cdot \frac{SE_r^2}{Ng}) = \tag{5}$$

$$\rho_g \sim N(\rho_B, SE_r^2) \tag{6}$$

Although the prior follows a normal distributions, in case of bounded parameters like correlations ($\rho \in [-1, 1]$), the prior is bounded by -1 and 1. Given that the prior is centered around a boundary value, for example $\rho_B = 0.1$, the area that agrees with H_i ($\rho \in [0.1, 1]$) is not always equal to the prior area that agrees with H_c ($\rho \in [-1, 0.1]$). In other words, depending on ρ_B, the complexity of H_i ($c(H_i)$) is not necessarily the same as that of H_c ($c(H_c)$).

In addition, when measurement error (SE) increases, the prior distribution becomes wider. Depending on ρ_B, the prior is then truncated asymmetrically. For example, when $\rho_B = 0.1$, a relatively larger proportion of the distribution is truncated at the upper bound (1) than at the lower bound (-1), leaving more mass in [-1,0.1] than in [0.1, 1]. As a result, the complexity ratio would shift in favour of H_c, potentially contributing to the observed shift in evidence as SE increases.

To test whether this asymmetric truncation of the prior could explain the observed pBF pattern, we examined whether the same shift occurs for effect sizes with unbounded parameter spaces ($es \in [-\infty, \infty]$), as truncation would not occur in this case.

Simulation 2: Covariances and Regression Coefficients (β)

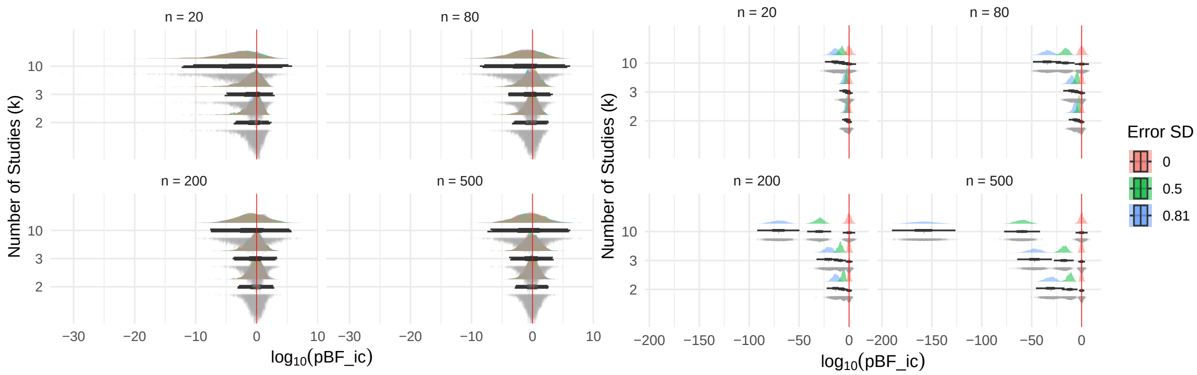
To answer the question how the pBF distributions behave when using effect sizes with unconstrained parameter spaces, we decided to include covariances, because these are unstandardised correlations, and regression coefficients (β), because these, like correlations, describe linear relationships. Again, four informed hypotheses were tested against their complement:

- (2a) H_i : cov / $\beta > 0.1$, when es $= 0.1$
- (2b) H_i : cov / $\beta > 0$, when es $= 0$
- (2c) H_i : cov / $\beta > 0.8$, when es $= 0.8$
- (2d) H_i : cov / $\beta > -0.8$, when es $= -0.8$

Because both covariance and β have unbounded parameter spaces, it was expected that for both effect size types, the $log_{10}\ pBF_{ic}$ distributions would be symmetrical around 0, indicating neutral evidence for both H_i and H_c.

As can be seen in Figure 3a (which shows simulation 2c because this is an extreme case), the covariances behaved exactly as we expected; in simulation 2a-2d, all $log_{10}\ pBF_{ic}$ distributions were nicely centered around 0 (all simulation plots can be found).

However, when β was the effect size of interest, the $log_{10}\ pBF_{ic}$ distributions did not behave as expected (see Figure 3b). All distributions showed the same pattern that was observed in



(a) True covariance = 0.8

(b) True beta = 0.8

Figure 3: Boundary case, Hi and Hc are both false. Red line: log(1) = 0, 1000 iterations, Number of studies (k), within study sample size (n) and reliability (errorsd)

the correlations, with more extreme boundary values resulting in stronger shifts towards the hypothesis that had the largest parameter space. This means that it is unlikely that the shift in evidence can be explained by prior truncation, as this does not happen in β.

Properties of the effect sizes

Aiming to understand why the shift in evidence was observed in ρ and β, but not in covariances, we decided to take a closer look at the data generating mechanism of the simulation and the properties of the effect size types.

In the data generating mechanism of this simulation, in the covariance structure that is assumed $Var(X) = Var(Y) = 1$.

$$\rho = Cor(X,Y) = \frac{Cov(X,Y)}{\sqrt{(}Var(X) \cdot Var(Y)} = Cov(X,Y) \tag{7}$$

Given that β is an ordinary least squares (OLS) estimator, the following holds:

$$\beta = \frac{Cov(X,Y)}{Var(X)} = Cov(X,Y) = \rho \tag{8}$$

This means that all effect sizes are constrained to be equal in the data generating mechanism. This raises the question why the pBFs of covariances behave differently then those of ρ and when $Cov(X,Y) = \beta = \rho$?

When we look at the data generating mechanism again, we have to take into account that noise is added to X and Y, such that the observed effect sizes are $X_{obs} = X + \epsilon$ and $Y_{obs} = Y + \epsilon$. Taking into account that ϵ is a random variable, translating this to the formulas of the effect sizes of interest, gives the following.

For the covariance:

$$
\begin{aligned}
&Cov(X_{obs}, Y_{obs}) = Cov(X + \epsilon_X, Y + \epsilon_Y) && (9)\\
&= Cov(X, Y) + Cov(X, \epsilon_Y) + Cov(Y, \epsilon_X) + Cov(\epsilon_X, \epsilon_Y) && (10)\\
&= Cov(X, Y) && (11)\\
&= \rho, && (12)
\end{aligned}
$$

given that ϵ_Y and ϵ_X are unrelated to each other and to X and Y.

For the correlation:

$$
\begin{aligned}
&\rho obs = Cor(X_{obs} + \epsilon_X, Y_{obs} + \epsilon_Y) && (13)\\
&= \frac{Cov(X_{obs}, Y_{obs})}{\sqrt{Var(X + \epsilon_X) \cdot Var(Y + \epsilon_Y)}} && (14)\\
&= \frac{\rho}{\sqrt{(Var(X) + Var(\epsilon_X)) \cdot (Var(Y) + Var(\epsilon_Y))}} && (15)\\
&= \frac{\rho}{\sqrt{(1 + errorsd^2) \cdot (1 + errorsd^2)}} && (16)\\
&= \frac{\rho}{1 + errorsd^2} && (17)
\end{aligned}
$$

For the regression coefficient:

$$
\begin{aligned}
&\beta_{obs} = \frac{Cov(X_{obs}, Y_{obs})}{Var(X_{obs})} && (18)\\
&= \frac{\rho}{(Var(X) + Var(\epsilon_X))} = \frac{\rho}{1 + errorsd^2} && (19)
\end{aligned}
$$

This means that both ρ_{obs} and β_{obs} are limited by the standard deviation that is specified in the model parameters, while this does not hold for the covariances. This process of attenuation may explain why the observed shift with increasing error is only visible in ρ and β, but not in the covariances. Since ρ_{obs} and β_{obs} are the effect sizes that are used to calculate BFs using bain, and both parameters shrink relative to their true values when there is error, the question becomes whether $\rho_{obs} > 0.1$, which it is not. This means that because of the attenuation, H_c in fact becomes true, and it is not surprising the pBF consistently provides evidence in favour of this hypothesis.

Simulation 3: Compare to RMA

To see if attenuation would also play a role using frequentist methods, RMA was used to pool correlation estimates from the k different studies. The mean correlation estimates and SEs resulting from this were then passed to `bain` to get one BF, which was repeated 1000 times, to then plot all BFs.

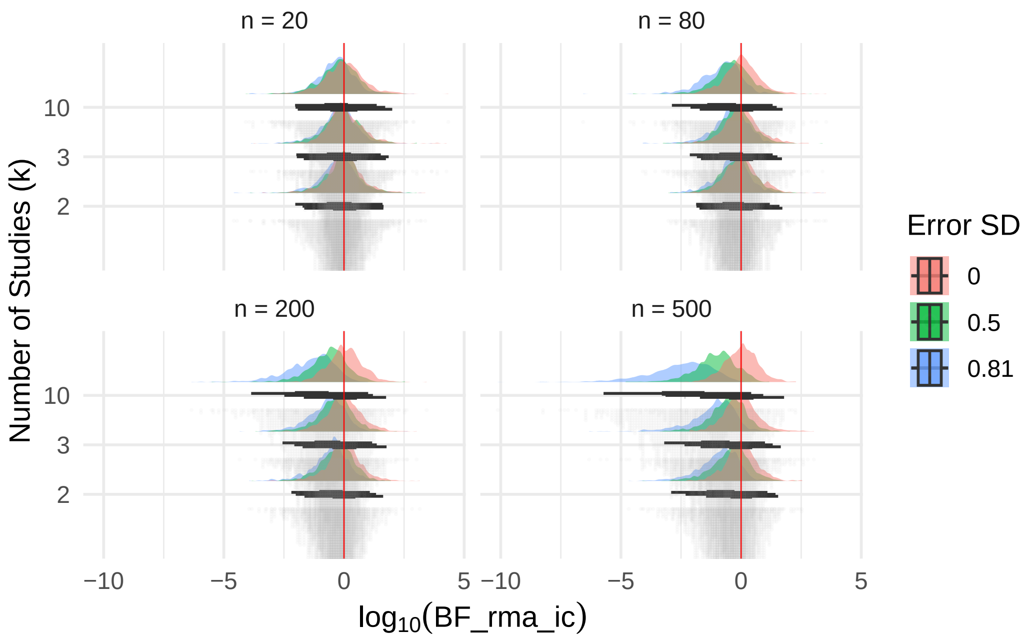


Figure 4: RMA, boundary case, true correlation = 0.1, Hi and Hc are both false. Red line: log(1) = 0, 1000 iterations, Number of studies (k), within study sample size (n) and reliability (errorsd)

Figure 4 shows the $log_{10}\ BF_{ic}$ distributions when first pooling all correlation estimates, and then performing RMA. It stands out that the shift in evidence towards H_c is also visible here. Consistent with the pBFs for ρ and β, with more extreme boundary values such as 0.8, the shift in BF values indicating evidence for H_c was more extreme. This means that the evidence shift introduced by measurement error persists at the level of the pooled estimate, and pooling correlation estimates via RMA before applying `bain` does not seem to protect against attenuation.

Simulation 4: Ordered hypotheses

Given that attenuation of correlation seems to play a role both in Bayesian and frequentist methods, the question remains how to deal with this. When testing ordered hypotheses, such as $H_i : \rho_1 < \rho_2 < \rho 3$, attenuation may not be as big of a problem. If all correlations have equal measurement error, they will all attenuate in the same manner, so it is expected that the order would remain the same as well. To see how the pBF behaves in this scenario, we set up a simulation where there were two conditions for *rho*:

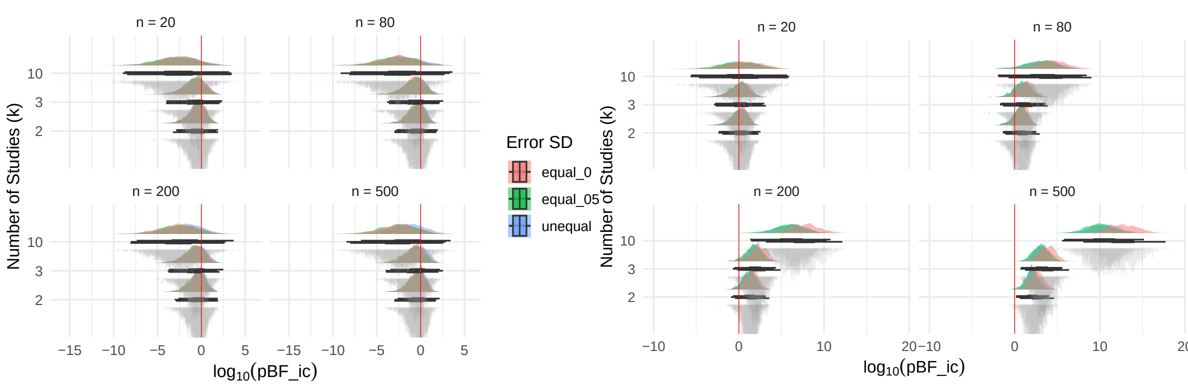
Table 2: Simulation conditions for ordered hypothesis Hi: r1 < r2 < r3.

Hypothesis	True Correlation	Error Condition
Hi is true	Ordered: r1 < r2 < r3	Equal at 0
Hi and Hc are both false	Equal: r1 = r2 = r3 = 0.1	Equal at 0.5
		Unequal: eX = 0.81, eY = 0, eZ = 0.5

- Unequal: ρ_1 = Cor(X,Y) = -0.1, ρ_2 = Cor(X,Z) = 0, ρ_3 = Cor(Y,Z) = 0.1, so H_i was true.
- Equal: $\rho_1 = \rho_2 = \rho_3 = 0.1$, so H_i and H_c were both false.

In addition, for the error, three different conditions were tested:

- Equal error at 0, to check if the pBF behaves as expected in the baseline condition when there is no error.
- Equal error at 0.5, to evaluate how the pBF behaves if all correlations attenuate with the same rate, it is expected this does not affect the pBF.
- Unequal error, where $\epsilon_X = 0.81$, $\epsilon_Y = 0$, $\epsilon_Z = 0.5$, to test whether having different measurement errors would affect the pBF.



(a) True r1 = -0.1, r2 = 0, r3 = 0.1, Hi is true

(b) True r1 = r2 = r3 = 0.1, Hi is false

Figure 5: Ordered hypothesis r1 < r2 < r3. Red line: log(1) = 0, 1000 iterations, Number of studies (k), within study sample size (n) and reliability (errorsd).

Figure 5a shows the $log_{10}\ BF_{ic}$ distributions where both H_i and H_c are false. It stands out that all distributions are centered around zero, even when there is error, suggesting that attenuation does not seem to affect the pBF when testing an ordered hypothesis. Only when the number of studies is large, there seems to be a small shift in evidence towards H_c.

In Figure 5b, two things become clear. First, when the within study sample size is small, it becomes clear that pBF does not increase the power to find an effect when it is indeed there. Secondly, regardless of whether all ρ values have equal or unequal error values, any error will induce a shift in evidence away from H_i.

In Figure 5b, it is clear that measurement error reduces the power of the pBF to detect a true ordered hypothesis. When errorsd = 0, the distributions shift rightward toward Hi, particularly with larger n and k. However, when measurement error is present - whether equal or unequal across groups - this shift is attenuated, with distributions remaining closer to zero. Importantly, measurement error does not appear to actively mislead the pBF toward Hc, suggesting that for ordered hypotheses, attenuation reduces sensitivity without inducing false evidence against Hi.

Figure 5b shows that when n is small (n = 20), the pBF distributions are centred around zero across all errorsd conditions, including when there is no measurement error (equal_0). This suggests that with small within-study sample sizes, the pBF lacks the power to detect the true ordering even when Hi is true. As n increases, the equal_0 condition shifts rightward, indicating that without measurement error the pBF does detect the true ordering with sufficient data. However, when measurement error is present (equal_05 and unequal), the distributions remain closer to zero regardless of n, suggesting that attenuation reduces the power to detect the true ordering. Notably, this reduction in power does not manifest as a shift toward HcH_c Hc - the pBF simply becomes less decisive rather than actively misleading.

When looking at the formulas, the fact that both error conditions that contain error greater than zero, it makes sense that the unequal and equal condition behave similarly.

For the correlation in the unequal condition:

$$\begin{aligned}
\rho_{1obs} &= Cor(X_{obs} + \epsilon_X, Y_{obs} + \epsilon_Y) && (20)\\
&= \frac{\rho_{1obs}}{\sqrt{(Var(X) + Var(\epsilon_X)) \cdot (Var(Y) + Var(\epsilon_Y))}} && (21)\\
&= \frac{\rho}{\sqrt{(1 + errorsd_X^2) \cdot (1 + errorsd_Y^2)}} && (22)
\end{aligned}$$

$$\begin{aligned}
\rho_{2obs} &= Cor(X_{obs} + \epsilon_X, Z_{obs} + \epsilon_Z) && (23)\\
&= \frac{\rho_{1obs}}{\sqrt{(Var(X) + Var(\epsilon_X)) \cdot (Var(Y) + Var(\epsilon_Y))}} && (24)\\
&= \frac{\rho}{\sqrt{(1 + errorsd_X^2) \cdot (1 + errorsd_Y^2)}} && (25)
\end{aligned}$$